Amogh Mannekote

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Research Interests

Evaluation of LLM-based simulations of human behavior · Generative agent-based modeling · LLM agent evaluation · Synthetic data generation

Education

University of Florida — Gainesville, FL
Ph.D. in Computer Science — Jan 2021 – Aug 2026
Dissertation: Neurosymbolic Authoring and Test-Time Compute Allocation for Generative Educational Simulations
Advisors: Dr. Kristy Elizabeth Boyer (chair), Dr. Bonnie J. Dorr (cochair)

Ramaiah Institute of Technology — Bengaluru, India
B.E. in Computer Science & Engineering — Aug 2014 – Aug 2018

Publications and Preprints

1. **Amogh Mannekote**, A. Davies, G. Li, K.E. Boyer, C. Zhai, B.J. Dorr, F. Pinto (2025). Do Role-Playing Agents Practice What They Preach? Belief-Behavior Consistency in LLM-Based Simulations of Human Trust. *Conference on Language Modeling (COLM 2025).*
2. **Amogh Mannekote**, A. Davies, J. Kang, K.E. Boyer (2025). Can LLMs Reliably Simulate Human Learner Actions? A Simulation Authoring Framework for Open-Ended Learning Environments. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025).*
3. **Amogh Mannekote**, J. Nam, Z. Li, J. Gao, K.E. Boyer, B. Dorr (2024). IndirectRequests: Making Task-Oriented Dialogue Datasets More Natural by Synthetically Generating Indirect User Requests. *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025).*
4. **Amogh Mannekote**, M. Celepkolu, J.B. Wiggins, K.E. Boyer (2023). Exploring Usability Issues in Instruction-Based and Schema-Based Authoring of Task-Oriented Dialogue Agents. *ACM Conference on Conversational User Interfaces (CUI '23).*
5. **Amogh Mannekote**, M. Celepkolu, A. Galdo, K.E. Boyer, M. Israel, S. Heckman, K. Stephens-Martinez (2022). Don't Just Paste Your Stacktrace: Shaping Discussion Forums in Introductory CS Courses. *SIGCSE 2022.*
6. X. Tian, **Amogh Mannekote**, C.E. Solomon et al. (2024). Examining LLM Prompting Strategies for Automatic Evaluation of Learner-Created Computational Artifacts. *Educational Data Mining (2024).*
7. **Amogh Mannekote**, B. Dorr, K.E. Boyer (2023). Agreement Tracking for Multi-Issue Negotiation Dialogues. (arXiv:2307.06524)
8. **Amogh Mannekote**, X. Tian, K.E. Boyer, B. Dorr (2024). Can Similarity-Based Domain-Ordering Reduce Catastrophic Forgetting for Intent Recognition? (arXiv:2402.14155)
9. **Amogh Mannekote** (2023). Towards a Neural Era in Dialogue Management for Collaboration: A Literature Survey. (arXiv:2307.09021)

Research Experience

University of Florida, *Graduate Research Assistant* — Gainesville, FL (Jan 2021 – Aug 2026)

- Built an evaluation harness for generative agent-based simulations of pair programming, with 50 behavioral targets across five families modeled on the forms that published pair-programming findings take, and joint reporting of target satisfaction, persona fidelity, and model-call cost.
- Designed two-stage verifiers that pair LLM extraction of 143 observable features with deterministic, rule-based scoring, so that presentation effects such as ordering and verbosity cannot affect scores; validated each target-verifier pair before use through feature-example checks, on-target versus off-target prompting audits, and manual review.
- Measured belief-behavior consistency in LLM role-playing agents, showing that models' stated predictions of persona behavior do not reliably match the behavior they simulate (COLM 2025).
- Showed that target locality (session-level vs. batch-level) predicts both target difficulty and which allocation strategy helps, with bootstrapped confidence intervals; batch-level targets require a planner, while verifier-guided search is most robust under the heaviest constraint load, at roughly 1.7x the inference cost.

- Reframed calibration of generative agent-based models as a test-time compute allocation problem, holding the generator and target specification fixed and varying only how inference-time compute is distributed across a batch.
- Implemented and benchmarked four inference-time allocation strategies (independent sampling, verifier-guided search, planner-guided generation, and persona-aware assignment) against a whole-batch baseline on GPT-OSS 120B and Nemotron 3 Super, across 50 behavioral targets at constraint loads of one to four.
- Showed with HYP-MIX that GPT-4 Turbo maintains behavioral calibration across 89% of compositional learner-model modifications (AAAI 2025).

Amazon Alexa, *Research Intern* Bellevue, WA (Sep 2024 – Dec 2024)

- Built MENTOR, a framework for measuring in-context adaptation to natural-language feedback and tool-use instructions on programming and reasoning tasks.
- Ran controlled studies isolating which feedback formats improve multi-step task performance, and characterized the failure modes that persist under them.

Openstream.ai, *Research Intern* Remote (Sep 2023 – Dec 2023)

- Built a LangChain-based evaluation pipeline for semantic parsing in a neuro-symbolic dialogue system, improving evaluation efficiency by 30%.
- Ran head-to-head comparisons of LLM-based semantic parsers with rubric-driven error analysis.

Amazon Alexa, *Research Intern* Bellevue, WA (May 2023 – Sep 2023)

- Designed and implemented a pipeline for synthetically generating indirect user requests in task-oriented dialogue using controlled linguistic attributes, increasing dataset diversity by 25%.
- Released the resulting dataset publicly (IndirectRequests, COLING 2025).

Fractal Analytics & Crux Intelligence, *Data Scientist* Bengaluru, India (Jan 2020 – Nov 2020)

- Built an end-to-end evaluation framework for Text-to-SQL semantic parsing, reducing release-cycle evaluation from one day to a few minutes.
- Automated regression checks and model-comparison workflows for a production-facing applied ML system.

Rasa, *Machine Learning Solutions Architect* Berlin, Germany (Jun 2019 – Nov 2019)

- Contributed to Rasa's open-source NLU and dialogue-management modules.
- Shipped conversational AI for enterprise clients in healthcare, automotive, insurance, and retail.

Earlier Roles

- **Indian Institute of Science**, *Research Assistant* (Jul 2018 – Jun 2019): benchmarked Visual SLAM approaches with sensor fusion in unstructured outdoor environments.
- **IBM**, *Intern, Watson Machine Learning* (Apr 2018 – Jun 2018): improved deployment efficiency for model-serving infrastructure.
- **Stride.AI**, *Research Intern* (Feb 2017 – Feb 2018): built a multi-turn conversational retrieval agent for employee skill matching.

Research Artifacts and Open-Source Projects

Generative Agent-Based Modeling Benchmark Research software for orchestrated pair-programming simulation, with scenario templates, behavioral constraints, evaluator families, and error-analysis tooling for studying interactional control in generative agent populations.

IndirectRequests LLM-plus-crowd data generation pipeline and public dataset of indirect user requests for schema-guided dialogue, built to stress-test dialogue understanding under more natural user language.

GPT-NegoChat Synthesized multi-issue negotiation dialogue corpus and agreement-tracking benchmark over a structured state space, with transfer-learning baselines built from MultiWOZ dialogue state tracking.

LLM4Qual Open-source library combining Hugging Face Datasets and LangChain to emulate human annotation workflows for qualitative text analysis in low-resource and privacy-sensitive research settings.

Nonechucks Open-source library extending PyTorch DataLoaders to handle invalid or unwanted samples in real-world ML pipelines (featured on GitHub Trending Python Repositories).

Skills

Programming Languages	Python, SQL, Java, TypeScript, JavaScript, Bash, LaTeX
ML & LLM Stack	PyTorch, TensorFlow, Hugging Face Transformers, LangChain, scikit-learn, Pandas, NumPy, Weights & Biases
Research Methodology	Hypothesis-driven experimentation, controlled studies, ablation studies, statistical analysis, reliability and robustness evaluation, error taxonomy design
Inference-Time Scaling	Test-time compute allocation, verifier-guided search, planner-guided generation, model-based verification, quality/compute tradeoff analysis
Agent-Based Modeling	Generative agent-based modeling, multi-agent simulation, simulation authoring, corpus-level calibration, belief-behavior consistency
NLP & Dialogue	Task-oriented dialogue, user simulation, semantic parsing, synthetic data generation, trajectory-level analysis
Systems & Compute	AWS, GCP, Docker, Kubernetes, SLURM, Git, Linux

Talks & Podcasts

Guest Speaker, CS-Ed Podcast	Discussion on the use of discussion forums in CS1/2 classrooms [Link]
Building a Chatbot with Rasa	Talk delivered at Ramaiah Institute of Technology [Link]

Academic Awards & Fellowships

Graduate Fellowship, 2022	Awarded by the Department of CISE, University of Florida
Student Travel Grant, 2019	Awarded for attending CoDS-COMAD 2019

Natural Languages

English (native) · Kannada (native) · Hindi (working)